STUDENT ID NUMBER _____

UNIVERSITY OF OTAGO EXAMINATIONS

COMPUTER SCIENCE

Paper COMP160

COMPUTER PROGRAMMING

Sample Final Exam No. 2

(TIME ALLOWED: TWO HOURS)

This examination paper consists of 16 pages including this cover page. **Please check that you have a complete set.** The total and maximum mark for this paper is 85.

Candidates **may** bring a single A4 sheet of notes (both sides) into the exam as reference material (no magnifying devices are allowed). Candidates **may not** bring in any other written material, calculators or electronic devices.

This exam is divided into multiple choice and short answer sections.

For the multiple choice questions:

Fill in your name and Student ID number on the form provided for your answers. Fill in the correct ovals for your Student ID. Put all answers on the form and use your general answer book for any rough work. Each multiple choice question is worth one mark. Select only one answer per question which you think is the BEST answer.

When marking your answers, **make sure you fill in the oval completely.** Remember your answer form is machine marked. Failure to complete the answer sheet correctly may result in lost marks. If you wish to change your choice, erase the original answer completely.

For the short answer questions:

Fill in your Student ID number on the top of each page of short answer questions in this script. Write all your answers in the spaces provided in this script, and use your general answer book for any rough work. Short answer questions are worth various marks as indicated, for example: (3). For any of these questions part marks may be given.

In this paper we have tried to represent some of the DrJava syntax colouring using **bold**, but this is for information only and mistakes in the use of bold or fonts should not be taken to be mistakes in the programs or script.

Please write your ID number in the top right hand corner of this page and the back page.

When you are finished hand in the answer sheet, this script, and your general answer book. All must have your ID number filled in.

TURN OVER

Multi-choice section. Please answer on the sheet provided.

1. What does the Java interpreter do?

   (A) It adds comments to Java programs.
   (B) It explains Java to new progammers.
   (C) It runs Java programs.
   (D) It translates Java into Swahili.
   (E) It filters coffee.

2. What do the Java libraries contain?

   (A) Packages of classes.
   (B) Classes of objects.
   (C) Containers of methods.
   (D) Instances of applets.
   (E) Collections of variables.

3. Which of the following statements about the method below is **not** correct?

```
public static List addTo (int i, double d) {  // method body  }
```

   (A) It has unrestricted visibility.
   (B) It returns a reference to an object of type List.
   (C) It is called "addTo".
   (D) The header specifies two formal parameters.
   (E) It is an instance member of its class.

4. Which of the following statements about data fields is **not** true?

   (A) Data fields can be used by any method in the class in which they are defined.
   (B) Data fields can contain primitive values or references.
   (C) Data fields can never be accessed outside the class in which they are defined.
   (D) Data fields that are declared as "final" cannot be modified.
   (E) If not explicitly initialised, data fields are initialised to "zero states".

5. Which of the following is a **correct** use of a name / identifier according to the Java naming conventions?

   (A) java.util is a class
   (B) Data is a method
   (C) greet() is a package
   (D) GST is a constant
   (E) while is a variable

6. Which of the following is **not** true of the Scanner class?

(A) It implements the Iterator interface.
(B) It has methods for processing tokens.
(C) Instances treat input data sources as streams.
(D) Instances can be used to write data to a file.
(E) Instances can be used to read data from the keyboard.

7. How do we express a condition or test within a program?

(A) Using a Unicode character.
(B) Using an assignment operator.
(C) Using a data field.
(D) Using a method call.
(E) Using a boolean expression.

8. Which of the following statements about local variables is **not** true?

(A) Local variables are defined within methods.
(B) Local variables cannot be declared as static.
(C) Local variables can be accessed only inside the block in which they are defined.
(D) Local variables may have a specified visibility.
(E) Local variables can be of primitive or reference types.

9. Which of the following statements is **false**?

(A) A mutator usually accepts input parameters.
(B) A constructor does not accept input parameters.
(C) An accessor usually includes a return statement.
(D) A constructor has the same name as the class it is defined in.
(E) A private data field can be accessed by mutators and accessors.

10. A program declares a data field as follows: `AClass ac = new AClass( );`
What kind of value does the data field `ac` hold?

(A) A class of type AClass.
(B) An object of type AClass.
(C) A reference to an object of type AClass.
(D) A reference to an array of type AClass.
(E) A reference to an array of references to objects of type AClass.

11. Which of the following is not achieved by the statement below?

```
int [ ] list = {3, 3, 2, 5, 1, 6, 7};
```

(A) It initialises the elements of the array referred to by the variable list.
(B) It initialises the variable list.
(C) It passes the elements of the array referred to by the variable list to a method.
(D) It declares a variable list of type array of int.
(E) The highest valid index of the variable list will be 6.

12. Given a string s with the value "Bilbo Baggins", which of the following will evaluate to the result "bo Baggins"?

(A) s.substring(s.indexOf('b'))
(B) s.indexOf(s.substring('b'))
(C) s.length(s.indexOf('b'))
(D) s.length() – s.substring('b')
(E) s.indexOf('b') – s.length()

13. A working program contains the code below. Which of the following is a different way to perform the same calculation of the value of b?

```
b = false;
    if (i <= 5)
        if ( i >= 0)
            b = true;
```

(A) b = i <= 5 || i >= 0;
(B) b = i <= 5 && i >= 0;
(C) b = !(i <= 5 && i >= 0);
(D) b = i <= 5, i >= 0;
(E) b = 5 >= i <= 0;

14. A for loop is supposed to process every element in an array data. Which of the following is the most appropriate header for the loop?

(A) **for** (**int** i = 0; i < data.length; i++)
(B) **for** (**int** i = 0; i <= data.length; i++)
(C) **for** (**int** i = 0; i == data.length; i++)
(D) **for** (**int** i = 1; i < data.length; i++)
(E) **for** (**int** i = 1; i <= data.length; i++)

15. In the main method of a working program (with a class AClass) a method is called as shown:

```
x = AClass.myMethod(2)
```

Which of the following statements about myMethod **is not required to be true**?

(A) myMethod is declared as static
(B) myMethod is not declared as private
(C) myMethod is declared as final
(D) myMethod is declared to take a single input parameter
(E) myMethod returns a result

16. Why is it useful to be able to have one class extend another?

(A) So that we can define new classes by adding to the functionality of existing classes instead of writing all classes from scratch.
(B) So that we can encapsulate the members of a class and restrict access other than by the use of accessor and mutator methods.
(C) So that we can create instance objects which share their private data fields.
(D) So that we can allow class objects to call each other's methods.
(E) So that we can organise classes into related packages instead of relying on the default package.

17. The values stored in the data fields of an object, and the methods of an object, are sometimes called by the terms:

(A) variables and parameters
(B) state and behaviour
(C) instance and class
(D) header and body
(E) cohesion and coupling

18. Which of the following statements about ArrayLists is false?

(A) An ArrayList is of fixed size.
(B) An ArrayList can only hold references to objects.
(C) ArrayList is one of the collection classes
(D) An ArrayList implements the Iterator interface.
(E) The ArrayList class is in the java.util package.

19. If Abs1 and Abs2 are abstract classes, and Inf1 and Inf2 are interfaces, which of the following is a legal header for the class Boris?

(A) **class** Boris **implements** Abs1 **extends** Inf1, Inf2 { *// class body* }
(B) **class** Boris **extends** Inf1 { *// class body* }
(C) **class** Boris **implements** Abs1, Abs2 { *// class body* }
(D) **class** Boris **extends** Abs1, Abs2 **implements** Inf1 { *// class body* }
(E) **class** Boris **extends** Abs1 **implements** Inf1, Inf2 { *// class body* }

20. What is the purpose of constructor chaining?

(A) To allow constructors in a parent class to access constructors in a child class.
(B) To allow class objects to be efficiently constructed.
(C) To prevent a selected constructor from executing.
(D) To make sure that every instance object is correctly constructed.
(E) To pass information from one constructor to another.

21. Which of the following is **not** true of inner classes?

(A) Only the class which contains them (the outer class) can make instances of them.
(B) Data fields in the outer class can contain primitive values or references.
(C) Data fields in the inner class can contain primitive values or references.
(D) Methods in the outer class can directly access members of the inner class.
(E) Methods in the inner class can directly access members of the outer class.

22. Which of the following are core features of the Java event model?

(A) buttons and checkboxes
(B) components and containers
(C) event generators and exception handling
(D) layout managers and abstract classes
(E) event classes and listener interfaces

23 When two or more methods in a class have the same name a different parameter list, this is an example of…

(A) overriding
(B) cohesion
(C) shadowing
(D) overloading
(E) coupling

24. Which of the following is the most useful way of approaching the design of a program that is to be controlled by a GUI?

(A) As a system of functions applied to data.
(B) As a logical formula whose correctness must be proved.
(C) As a specification of how to respond to user inputs.
(D) As a system of interacting automatic processes.
(E) As a specification of a sequence of operations.

Short answer section.

Please fill in your ID number on the top of all short answer pages, and write answers in the spaces provided in this script.

1. What are the two basic kinds of objects in Java? (1)

   Class Objects _____ Instance Objects _____

2. What is the general term for errors that are detected by the compiler? Syntax Error _____ (1)

   What is the general term for errors that are detected by the interpreter? Run Time Error _____

3. What package of classes is automatically imported into every Java program? (1)

   Java.lang _____

4. Mathematical constants are implemented using Data Fields _____, (1)

   and mathematical functions are implemented using Methods _____

   in the class Math.

5. For **each part** of this question assume the following variables and the initial values shown.

```
int i = 5, j = 8;
double d = 2.1;
char c = 'x';
boolean b1 = true, b2 = false;
```

   Show what each expression evaluates to:

i *= j	40	j / (2 / 2)	8	(1)
–5 + 2 * 3 + 1	2	(double) i	5.0	(1)
++c	'y'	b1 && !b2	true	(1)
j % 5	3	j / 5	1	(1)

6. Write boolean expressions to test the following conditions: (1)

   a char value ch is a space if (ch.equals(" "))

   an int value i which is outside the range 12 to 18 if (i < 12 && i > 18)

7. Write a method called tester which accepts two integer values as input parameters and returns the boolean result true if the sum of the inputs is greater than or equal to 10 (and false otherwise). (2)

```
public boolean tester(int x, int y){
   if ((x + y) >= 10){
     return true;
   }
   else{
     return false;
   }
 }
```

8. Using an "if else if" structure write a sequence of conditions so that: if an integer variable called sum is positive the output "positive" is printed, if it is zero then the output "zero" is printed, and otherwise the output "negative" is printed. (2)

```
if (sum == 0){
     System.out.println("zero");
   }
   else if (sum > 0){
     System.out.println("positive");
   }
   else{
     System.out.println("negative");
   }
```

9. Assume a readInt method (as used in lectures and labs) that reads integers from the user. Write a while loop that reads in integers and each time writes out the sum of the values read so far. Continue until the sum exceeds 100 (i.e. the final sum written will be the first that exceeds 100). Include any declarations and initialisations required. (2)

```
import java.util.Scanner;

Scanner scan = new Scanner(System.in);
int sum = 0;
while(sum <= 100){
    sum += scan.nextInt();
    System.out.println(sum);
}
```

10. Write a class Book which has two data fields, int pages (which represents the number of pages) and String title (which represents the title of the book). Write a constructor which uses two input parameters to initialise the data fields, and a constructor which replaces the default constructor. (3)

Write a second class, TextBook, which extends Book, and has a data field cost of type double (which represents the cost of the book). Write a constructor which uses three input parameters to initialise pages, title and cost, but does so by using a constructor in the parent class where possible. (3)

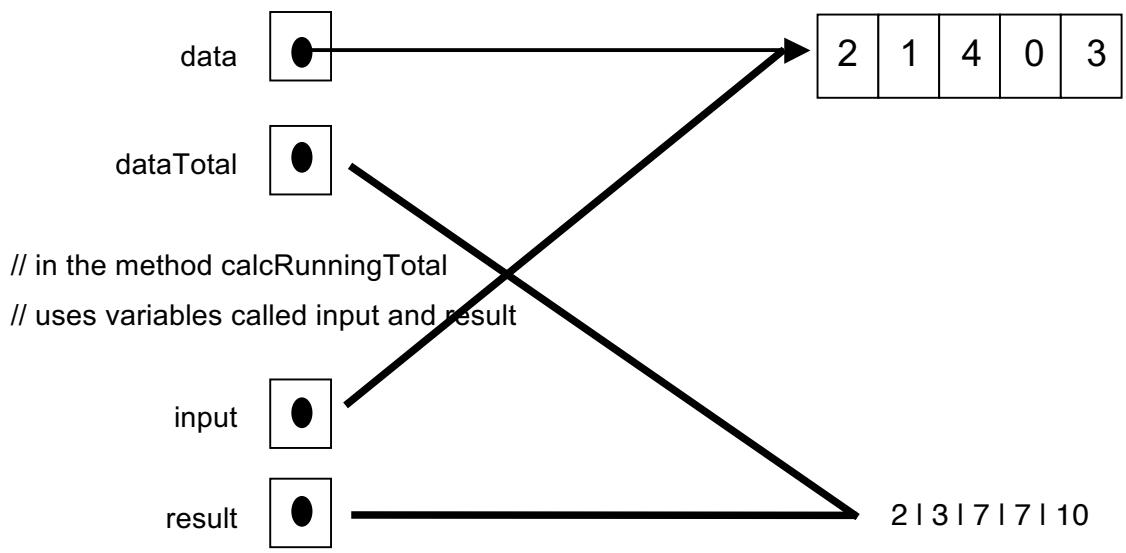
```
public class Book{
  protected int pages;
  protected String title;

  public Book(){}

  public Book(int i, String s){
    pages = i;
    title = s;
  }
}



public class TextBook extends Book{
  protected double cost;

  public TextBook(){}

  public TextBook(double d, int i, String s){
    super(i,s);
    cost = d;
  }
}
```

11. Write a static method calcRunningTotal which has a formal parameter called input of type array of int (so that this variable can refer to an array such as for example: {2, 1, 4, 0, 3}). The method should make a new array result of the same size, and use a loop to set the elements of result to contain a running total of the elements of input (for the example above result would hold: {2, 3, 7, 7, 10}). The method should return a reference to result. (4)

```
public static int[] calcRunningTotal(int[] input){
   int sum = 0;
   int[] result = new int[input.length];
   for (int i = 0; i < input.length; i++){
     sum += input[i];
     result[i] = sum;
   }
   return result;
 }
```

12. If the method calcRunningTotal described above is called (from the main method) as shown, add to the diagram any further array objects that are necessary, and the three remaining arrows representing the references stored in the four variables. (2)

```
// in the main method
int [ ] data = {2, 1, 4, 0, 3};
int [ ] dataTotal = calcRunningTotal(data);
```



13. Given the classes below, assume that b refers to an object of type B. What six lines of output are created when we call b.test() ? (3)

2

4

3

B here

B here

A here

```
public class A {
        int x = 2;        // line 1

        public void aMethod( ) {        // line 2
                System.out.println("A here");
        }
}

public class B extends A {
        public int x = 3;        // line 3

        public void aMethod( ) {        // line 4
                System.out.println("B here");
        }

        public void test( ) {
                int x = 4;
                System.out.println(super.x);
                System.out.println(x);
                System.out.println(this.x);
                this.aMethod();
                aMethod();
                super.aMethod();
        }
}
```

What is the relationship between the lines labeled *line 1* and *line 3* ? (1)

both int called x

What is the relationship between the lines labeled *line 2* and *line 4* ? (1)

both public void methods called aMethod, method overloading

14. What three lines of output are written when this program runs? (3)

0.0

3.0 12

4.0 8

```
public class MyApp {

    public static void main(String[ ] args) {
        System.out.println(Data.s);
        Data dat1 = new Data(3.0, 12);
        Data dat2 = new Data(4.0, 8);
        System.out.println(dat1);
        System.out.println(dat2);
    }

}

public class Data {

    public static double s;
    public int i = 2;

    public Data(double s, int i) {
        this.s = s;
        this.i = i;
    }

    public String toString() {
        return s + " " + i;
    }

}
```

15. At the root of the Java class hierarchy tree is the class Object. (1)

16. Given the following classes: (2)

Animal

Mammal **extends** Animal

Dog **extends** Mammal

Spot **extends** Dog

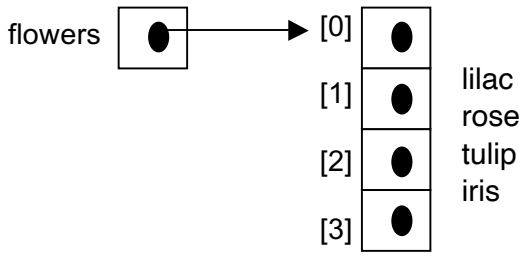
What are **all** the legal types of an object which is an instance of Dog?

Animal, Mammal, Dog

17. A working program contains the following code:

```
ArrayList<String> flowers = new ArrayList<String>();
flowers.add("rose");
flowers.add("tulip");
flowers.add("iris");
flowers.add(1, "rata");
flowers.add(0, "lilac");
flowers.remove(2);
```

Complete the model sketch below to show the state of the ArrayList when this code has executed. (2)



18. Write a foreach loop which will print out every String stored in the ArrayList flowers from the previous question. (1)

```
for (String s: flowers){
    System.out.println(s);
}
```

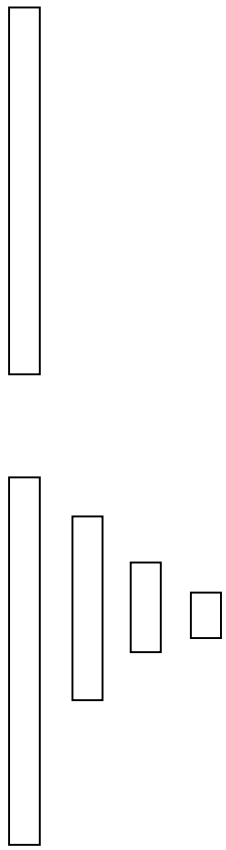
19. An application class makes an instance of this JPanel and displays it in a JFrame (window). The paint method draws rectangles, with the top left corner at the position specified by top and left, and the width and height specified by width and height.

As currently shown the paint method draws 4 rectangles in the same place, producing a single image of a rectangle (as shown in the top output).

Add the import statements required to make the class compile. Make the three additions necessary so that when the program is run it produces each rectangle half the height of the previous one, and with the top left corner moved by 20 pixels in the x and y dimensions (as shown in the bottom output).

(3)

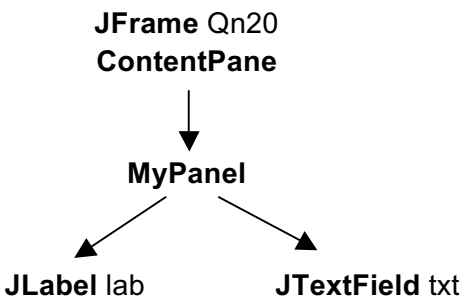
```
/* Anthony, October 2008, graphics demo */

    import javax.swing.*;                          // import
    ____________________________________________
      import java.awt.*;                           // import
    ____________________________________________

public class SillyRectangles extends JPanel {

      public void paint (Graphics g) {

        int top = 50, left = 50, width = 10, height = 200;

        for (int i = 1; i <= 4; i++) {

                g.drawRect (top, left, width, height);

                g.drawRect(top+20, left+20, width, height/2);
                ____________________

                g.drawRect(top+40, left+40, width, height/4);
                ____________________

                g.drawRect(top+60, left+60, width, height/8);
                ____________________

        }

    }

} // Class SillyRectangles
```

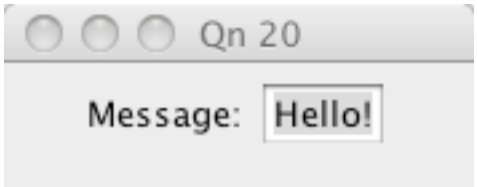


Give two further examples of methods that can be called on Graphics objects. (1)

g.drawString("Hello",0,0); g.fillOval(20,20,20,20);

20. An application class makes an instance of the class MyPanel (which extends JPanel), and displays it in a JFrame (window) as shown below. The full containment hierarchy is also shown below. (3)

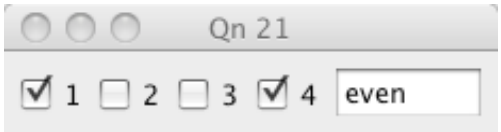


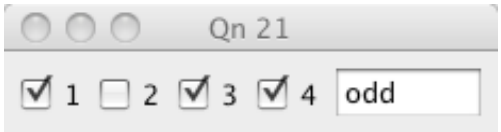


Write the constructor for the class MyPanel. Use the variable names, lab and txt as shown. Note that the strings displayed in the JLabel ("Message:") and JTextField ("Hello!") can be set by passing them as parameters to the constructors for these objects.

```
private JLabel lab;
private JTextField txt;

public MyPanel(String lab, String txt){
    this.lab = new JLabel(lab);
    this.txt = new JTextField(txt);
    add(this.lab);
    add(this.txt);
}
```

21. An application class makes an instance of BoxPanel (on the next page) and displays it in a JFrame (window). Whenever the user clicks on a check box the text field updates to show whether the number of checkboxes currently selected is odd or even. A couple of sample states are shown below.





Add the missing code indicated in six places to complete the program. (6)

Notes: The ItemListener interface specifies an itemStateChanged method. Event generators have an addItemListener method for registering listener objects.

```
/* Anthony, October 2008, Odd or even check boxes */

import javax.swing.*;   import java.awt.event.*;   import java.awt.*;

public class BoxPanel extends JPanel {

    private JTextField display = new JTextField("", 5);
    private JCheckBox [ ] boxes = {  new JCheckBox(), new JCheckBox(),
        new JCheckBox(), new JCheckBox()
    };

    public BoxPanel() {
        TheListener listener = new TheListener();                    // make a listener

        int i = 0;
        for (JCheckBox b: boxes)                                     { // foreach check box b
            i++;
            b.setText("" + i);
            b.addItemListener(listener);                             // register the listener
            add(b);
        }
        add(display);

    } // end of constructor

    private class TheListener implements ItemListener {

        public void actionPerformed(ItemEvent e)                     {
            int count = 0;
            for(JCheckBox b : boxes) {
                if  (  b.isSelected()                                ) {
                    count ++;                                        // update count
                }
            }
            if (  (count % 2) == 0                                   ) {
                display.setText("even");                             // display text
            } else {
                display.setText("odd");                              // display text
            }
        }  // end of method

    } // end of TheListener class

}   // end of BoxPanel class
```